

Miscellaneous Algebraic Knowledge

Explain: How we set our tests and select our team, purpose of training, may talk about University stuff, no need to be restricted by syllabus.

Important General Things:

Idea (Concepts, definitions, approaches)

Scope (Depth, Range, complexity)

Machinery (Technique, theory, story)

Maturity (Read Intensively, Journal, chat, practice)

Appreciation (Even you can't create, appreciate)

Novel (Read like a novel)

No Mystery. Try to understand.

(In short, knowledge, skill, insight, importance not necessarily in that order.)

Algebra, other areas: **Geometry, Analysis, Topology**. (**Combinatorics is not even in the list.**)

Coordination, Quantification (Shafarevich)

Structure

Representation

Character

Build Up Number Systems, Integer, Rational, Irrational, Real. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.

Vectors, matrices (operations), vector products (dot and cross), multiplication, linear systems, Gauss Elimination, determinants, Cramer's rule, vector spaces, basis. $\mathbb{R}^2, \mathbb{R}^3, \mathbb{R}^n, i, j, k, \mathbb{C}^2$. Ways to find determinant, inverse of 2×2 matrices, $n \times n$ matrices, eigenvalues and eigenvectors, Cayley-Hamilton's Theorem.

Miscellaneous Algebraic Systems, Group (Lagrange), permutation groups, symmetry group of cube, Ring (CRT homomorphism, isomorphism, domain), Field, Vector (Space), Matrix, Finite Field, Coding Theory, Polynomials. Wedderburn's Theorem on finite division ring.

TAKE THREE HOURS TO FINISH THE ABOVE.

Complex, Quaternions.

Complex Numbers, representation, Cartesian, polar, exponential, application (geometry and formula, most important, norm and geometric properties related to products of complex numbers), extension to 3-D, Hamilton story, Introduction to quaternions, unit quaternions, rotation, isomorphism. Illustration of how to define e^{iy} by expanding using Taylor series or by setting $e^{iy} = \cos y + i \sin y$. Complex numbers isomorphic to real 2×2 matrices of the form $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$. (Verify).

16th, October, 1843, mathematical vandalism, Brougham Bridge, Dublin, by W. R. Hamilton.

$$i^2 = j^2 = k^2 = ijk = -1.$$

Motivation

Try to multiply elements of the form $a + bi + cj$, with $i^2 = j^2 = -1$, and i, j independent norm 1 vectors.

Ideas:

(i) Assume $ij = ji$, then

$$\begin{aligned} -1 = i^2 = j^2 &\Rightarrow 0 = i^2 - j^2 = i^2 - ij + ji - j^2 \\ &= i(i - j) + j(i - j) = (i + j)(i - j) \Rightarrow i = \pm j, \\ &\text{contradiction. Hence } ij \neq ji. \end{aligned}$$

(ii) Assume $a + bi + cj$ a norm 1 vector, then

$$\begin{aligned} 1 &= a^2 + b^2 + c^2 = (a + bi + cj)(a - bi - cj) \\ &= a^2 + b^2 + c^2 - bc(ij + ji), \end{aligned}$$

hence should have $ij = -ji$.

(iii) Now what is ij ? Try

$$ij = a + bi + cj, \text{ get}$$

$$\begin{aligned} -i &= ij^2 = (a + bi + cj)j \\ &= aj + bij - c = aj + b(a + bi + cj) - c \\ &= aj + ba + b^2i + bcj - c \\ &= (c + ba) + b^2i + (a + bc)j. \end{aligned}$$

Hence $b^2 = -1$, a contradiction!

Example (Exercise). Using the definition of Hamilton, get $ij = k, kj = i, ji = k$. Also $ji = -k, jk = -i, ik = -j$.

Representations

A quaternion $a + bi + cj + dk$, if the situation is clear, is also usually written as (a, b, c, d) .

Theorem. Sending a quaternion (a, b, c, d) to the 2×2 complex matrix $\begin{pmatrix} a + bi & -c - di \\ c - di & a - bi \end{pmatrix}$ is an $\mathbb{R}$ -algebra isomorphism.

Likewise the real quaternions form an algebra which is isomorphic to the 4×4 real matrices:

$$\begin{pmatrix} a & -b & -c & -d \\ b & a & -d & c \\ c & d & a & -b \\ d & c & b & a \end{pmatrix}.$$

Conjugate, norm of a quaternion, finding inverse.

The *conjugate* of $q = a + bi + cj + dk$ is $\bar{q} = a - bi - cj - dk$. Its *norm* is

$|q| = \sqrt{a^2 + b^2 + c^2 + d^2}$. Check that $q\bar{q} = |q|^2$. Note the inverse of q is $\frac{\bar{q}}{|q|^2}$.

Exercise. $\overline{q_1 q_2} = \bar{q}_2 \bar{q}_1$.

Another form of writing a quaternion and a product of two of them.

Another way of writing a quaternion $a + bi + cj + dk$ as (a, u) , where a is the same and u stands for the vector $bi + cj + dk$ (3-vector). Adding is the same component-wise, and product

$$(a_1, u_1)(a_2, u_2) = (a_1 a_2 - u_1 \cdot u_2, a_1 u_2 + a_2 u_1 + u_1 \times u_2).$$

Note the appearances of the dot and cross product.

Exercise. Check consistent with the original definition.

Unit quaternions and theorem about rotation. A quaternion $a + bi + cj + dk$ is a *unit quaternion* if $|q| = \sqrt{a^2 + b^2 + c^2 + d^2} = 1$.

Theorem: For any unit quaternion $q = \cos \frac{\theta}{2} + \sin \frac{\theta}{2} u$, and for any $v \in \mathbb{R}^3$, the action of the operator $L_q(v) = qvq^{-1}$ on v is equivalent to the rotation of v through an angle θ about u as the axis of rotation. Here v is regarded as the quaternion as $(0, v)$.

Exercise. (i) With the notation above, check $L_q(u) = u$. (ii) What is L_{-q} ? (iii) What is $L_{\cos \frac{\theta}{2} - \sin \frac{\theta}{2} u}$? ((i) Easy to check, (ii) Same, (iii) Inverse).

Euler Rotation Formula: Composition of rotations (with the centre always fixed) is a rotation. (Every rotation in an odd-dimensional space has a fixed axis).

A unit quaternion and its variations.

Example. First turn 90° about the z -axis, then 90° about the x -axis. What is the resulting rotation?

Turning 90° about the z -axis corresponds to the unit quaternion $q_1 = \cos 45^\circ + \sin 45^\circ (0i + 0j + k)$. Turning about the x -axis corresponds to the unit

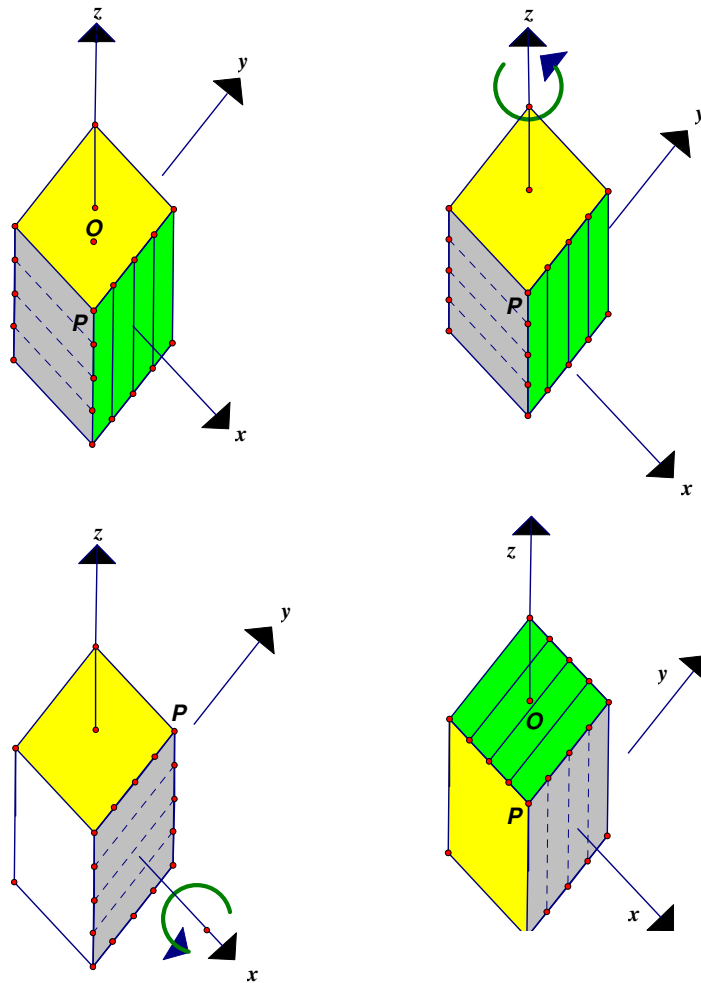
quaternion $q_2 = \cos 45^\circ + \sin 45^\circ (i + 0j + 0k)$. Multiplying

$$\begin{aligned} q_2 q_1 &= \frac{1}{2} + \frac{1}{2}i - \frac{1}{2}j + \frac{1}{2}k = \frac{1}{2} + \frac{\sqrt{3}}{2} \left(\frac{1}{\sqrt{3}}i - \frac{1}{\sqrt{3}}j + \frac{1}{\sqrt{3}}k \right) \\ &= \cos 60^\circ + \sin 60^\circ \left(\frac{1}{\sqrt{3}}i - \frac{1}{\sqrt{3}}j + \frac{1}{\sqrt{3}}k \right). \end{aligned}$$

Hence it is a rotation corresponds to turning about the axis $\frac{1}{\sqrt{3}}i - \frac{1}{\sqrt{3}}j + \frac{1}{\sqrt{3}}k$

(120°).

The pictures illustrate the rotations.



(Note O is in the centre of the cube.)

Example. (i) Determine the image of the point $(1, -1, 2)$ under the rotation of an angle 60° about an axis in the yz -plane that is inclined at an angle of 60° to the positive y -axis. (ii) Now consider the rotation of 60° about an axis in the xz -plane inclined at an angle to the positive x -axis. Determine the composition of the rotation in (i), the one in (i) acts first.

(i) The unit vector in the axis of rotation is $\cos 60^\circ j + \sin 60^\circ k$, hence the corresponding unit quaternion is

$$\begin{aligned} \cos 30^\circ + \sin 30^\circ (\cos 60^\circ j + \sin 60^\circ k) &= \frac{\sqrt{3}}{2} + \frac{1}{2} \left(\frac{1}{2} j + \frac{\sqrt{3}}{2} k \right) \\ &= \frac{1}{4} (2\sqrt{3} + j + \sqrt{3}k), \end{aligned}$$

$$\text{with its inverse } q^{-1} = \frac{1}{4} (2\sqrt{3} - j - \sqrt{3}k).$$

The corresponding image of $(1, -1, 2)$ is therefore

$$q(1, -1, 2)q^{-1} = q(0 + i - j + 2k)q^{-1} = \frac{1}{8} ((10 + 4\sqrt{3})i + (1 + 2\sqrt{3})j + (14 - 3\sqrt{3})k).$$

(ii) The new unit vector of rotation is $\cos 60^\circ i + \sin 60^\circ k$, hence the new unit quaternion is

$$\begin{aligned} \cos 30^\circ + \sin 30^\circ (\cos 60^\circ i + \sin 60^\circ k) &= \frac{\sqrt{3}}{2} + \frac{1}{2} \left(\frac{1}{2} i + \frac{\sqrt{3}}{2} k \right) \\ &= \frac{1}{4} (2\sqrt{3} + i + \sqrt{3}k). \end{aligned}$$

We then compute

$$\begin{aligned} \frac{1}{4} (2\sqrt{3} + i + \sqrt{3}k) \cdot \frac{1}{4} (2\sqrt{3} + j + \sqrt{3}k) &= \frac{1}{16} (9 + \sqrt{3}i + \sqrt{3}j + 13k) \\ \frac{9}{16} + \frac{1}{16} \cdot 5\sqrt{7} \left(\frac{\sqrt{3}i + \sqrt{3}j + 13k}{5\sqrt{7}} \right) &= \frac{9}{16} + \frac{5\sqrt{7}}{16} \left(\frac{\sqrt{21}i + \sqrt{21}j + 13\sqrt{7}k}{35} \right). \end{aligned}$$

$$\text{The corresponding rotation angle is } \cos^{-1} \frac{9}{16} \approx 55.77^\circ.$$

Topology The fundamental group of $SU(2)$ is trivial, fundamental group of $SO(3)$ is $\mathbb{Z}_2$.

TAKE THREE HOURS TO TALK ABOUT COMPLEX NUMBERS AND QUATERNIONS etc., HARD TO DISCUSS FUNDAMENTAL GROUPS.

Exercise

1. The subset $\{\pm 1, \pm i, \pm j, \pm k\}$ forms a group.
2. Quaternions with rational coefficients form a division ring.
3. The equation $x^2 = -1$ has infinitely many quaternion solutions. (Use different forms of quaternions.)
4. Verify the set I of quaternions with coefficients either are integers or halves of odd integers is a subring.
5. (Hua) Let a and b be elements of a ring such that a , b and $ab-1$ are units. Show that $a-b^{-1}$ and $(a-b^{-1})^{-1}-a^{-1}$ are units and the following identity holds:

$$((a-b^{-1})^{-1}-a^{-1})^{-1} = aba - a.$$

(For the identity, first check $(a-b^{-1})^{-1} = b(ab-1)^{-1}$ by multiplying $a-b^{-1}$ to the right-hand side, then apply it to the identity.)

6. (Hua's identity again) In a division ring, if $ab \neq ba$, then

$$a = (b^{-1} - (a-1)^{-1}b^{-1}(a-1))(a^{-1}b^{-1}a - (a-1)^{-1}b^{-1}(a-1))^{-1}.$$

(Just multiply a to $(a^{-1}b^{-1}a - (a-1)^{-1}b^{-1}(a-1))$ and simplify.)

7. (Cartan-Brauer-Hua) Let D be a division ring, C its centre and let S be a division subring of D which is stabilized by every map $x \rightarrow dxd^{-1}, d \neq 0$ in D . Show that either $S = D$ or $S \subseteq C$. (Try to use 6. First from 6, every division ring is generated by a non-central element and its conjugates, and the statement follows.)
8. Find the centre of all real quaternions. (The real subfield. Let $C_i = \mathbb{R}[i], C_j = \mathbb{R}[j], C_k = \mathbb{R}[k]$. Check the centre $Z_H(C_i) = C_i$ and so on. As

$$H = C_i + C_j + C_k, \text{ then } Z(H) = Z(C_i) \cap Z(C_j) \cap Z(C_k) = \mathbb{R}.)$$

Many other Simple and Important Results (Chinese Remainder Theorem, Lagrange' Theorem, Cyclotomic Fields, Minimal Polynomials, Primitives, Quadratic Reciprocity, examples.

References

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